One rotor in three gravitational fields

What the clocks, spin axis and a light loop of a fast rotor show in Earth's field, in orbit and near a neutron star, and how much the rotor itself curves spacetime. CLAUDE.md §3l

kronspice simulation report · 2026-11-07 · T1000G disk, 0.5 m radius, at half its burst speed

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Summary

A 0.5 m carbon-fibre (Toray T1000G) disk, 2 cm thick, spins at half its burst speed and carries a 1 m² fibre loop. The example combines rotation (special relativity) and curvature (general relativity) for this one machine at three sites, using the *spacetime/* package, and then asks the reverse question: how much does the machine itself curve and twist spacetime? It is a standard-physics example; the one hypothesis on this bench, the rotor energy-shift anomaly H17, is computed in *spacetime.RotorShift* with the standard k = 1/2 beside a declared k.

- **The rim loses 4.25×10^{-7} s/day (0.425 µs/day) against the hub at every site**, because the lag depends only on the rim speed (941 m/s); the hub rate carries the gravity.
- **Gravity Probe B check:** averaged over the 642 km polar orbit the model gives 6621.0 mas/yr geodetic and 40.85 mas/yr frame dragging, 39.1 mas/yr of it seen against GP-B's guide star; measured 6601.8 ± 18.3 and 37.2 ± 7.2 mas/yr.
- **The material limits the rotor first:** burst margin 2.00 at every site; at the burst speed $(v/c)^2$ is 4×10^{-11}.
- **Near the illustrative neutron star** the hub clock runs at 0.997930583704828 (GM/rc^2 = 0.00207, a strong field) and frame dragging reaches 4.87×10^{11} mas/yr; tidal stress 3.34×10^{5} Pa still leaves a margin of 1.9×10^{4}.
- **The machine's own spacetime is tiny:** 1 m away its clock shift is 2.1×10^{-26} (3×10^{-17} of Earth's) and its frame dragging 6.43×10^{-8} mas/yr (3.7×10^{-10} of Earth's at 45°); only its tidal pull (0.0012 of Earth's) is within reach of torsion balances.
- **No energy enters a circuit from spacetime curvature:** the gravitational channel carries clock rates and frames, not power.

The construction

CLAUDE.md §3l keeps three "curvatures" apart; this example uses the first two.

- **Gravitational curvature.** The hub sits in the Schwarzschild field of the body at its site, moving with the site: clock rate $\sqrt{(1 - r_s/r - v^2/c^2)}$ (r_s the Schwarzschild radius, r the distance, v the site's speed), exact form in strong fields, with tidal stretching and the horizon and photon-sphere limits as typed errors.
- **Rotation in flat spacetime.** In the hub's local inertial frame the rim is ordinary special relativity (the Langevin metric): rim clocks run at $\sqrt{(1 - \omega^2r^2/c^2)}$, a loop on the rotor shows the Sagnac delay $4A\omega/c^2$ (A the loop area, ω the spin rate), and the rim speed is bounded by the material's burst speed $\sqrt{(\sigma/\rho)}$.
- By the equivalence principle the two compose: the exact rim rate equals the first-order weak-field metric's rate to second order in the small quantities. The spin axis precesses by Thomas + geodetic + Lense–Thirring (Schiff 1960); the loop's Sagnac delay is taken relative to the local gyroscopes, with a "local-frame correction" for the local frame's precession.

The rotor

Toray T1000G fibre, 6,370 MPa and 1.80 g/cm³, bursts as a thin hoop at $\sqrt{(\sigma/\rho)}$ = 1,881 m/s (a solid disk bursts lower; the fibre value is an upper bound for a composite rotor). The example runs at half that, 941 m/s at 0.5 m, so ω = 1,881 rad/s. The disk's mass is $1800\cdot\pi\cdot0.5^2\cdot0.02$ = 28.3 kg and its angular momentum $\frac{1}{2}MR^2\omega = 6.65\times10^{3}$ kg·m²/s.

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The three sites

site	body	position and motion
Earth laboratory	Earth (GM, J2-free, rotating)	surface at 45° latitude, moving with the ground
Gravity Probe B orbit	Earth	circular polar orbit at 642 km above the equatorial radius, inertially pointed
neutron star (illustrative)	1.4 solar masses, 12 km radius, 10 ms spin, I = 0.4·M·R² (assumed values, not a measured object)	hovering 1,000 km from the centre, held up by a rocket

Solar GM is the IAU 2015 Resolution B3 nominal value.

Results

The rotor at each site

	Earth lab, 45°	GP-B orbit, 642 km	neutron star, 1,000 km (illustrative)
hub clock rate	0.999999999304050	0.999999999052363	0.997930583704828
rim clock rate	0.999999999299128	0.999999999047441	0.997930583699917
rim loses against hub	4.25×10^{-7} s/day	4.25×10^{-7} s/day	4.25×10^{-7} s/day
GM/rc²	6.95×10^{-10}	6.32×10^{-10}	0.00207
v²/c² at the rim	9.84×10^{-12}	9.84×10^{-12}	9.84×10^{-12}
burst margin	2.00	2.00	2.00
tidal stress on the disk (margin)	0.00277 Pa (2.3×10^{12})	0.00207 Pa (3.07×10^{12})	3.34×10^{5} Pa (1.9×10^{4})
Thomas precession, here	116 mas/yr	0	0
geodetic precession, here	350 mas/yr	6.62×10^{3} mas/yr	0
frame dragging, here	172 mas/yr	81.7 mas/yr	4.87×10^{11} mas/yr
Sagnac on the 1 m² loop	8.372×10^{-14} s per pass	8.372×10^{-14} s	8.372×10^{-14} s
local-frame correction to Sagnac	-1.05×10^{-30} s	4.53×10^{-29} s	0 s

- Averaged over the GP-B orbit the model gives 6621.0 mas/yr geodetic and 40.85 mas/yr frame dragging along Earth's axis, of which 39.1 mas/yr is seen against the mission's guide star IM Pegasi (declination +16°50′; GP-B measured the component perpendicular to the star line). Measured: 6601.8 ± 18.3 and 37.2 ± 7.2 mas/yr.
- At the neutron star the program warns that GM/rc² = 0.00207 is a strong field: the clock rates use the exact Schwarzschild form, and the precession and Sagnac terms are first-order and only indicative.

The machine's own spacetime

1 m from the 28.3 kg rotor (J = 6.65×10^{3} kg·m²/s), on its axis:

	rotor	Earth (surface, or at 45°)	ratio
clock shift GM/dc²	2.1×10^{-26}	6.95×10^{-10}	3×10^{-17}
tidal curvature GM/c²d³	2.1×10^{-26} /m²	1.71×10^{-23} /m²	0.0012
frame dragging on the axis	9.87×10^{-24} rad/s = 6.43×10^{-8} mas/yr	172 mas/yr	3.7×10^{-10}

A machine changes curvature through its mass and energy, and its spin drags frames, but at laboratory scale the tidal curvature is the only part within reach (torsion balances weigh that pull), and the frame dragging is ten orders below Earth's own. The rotor's rim speed is limited by its material long before any relativistic effect becomes large.

Validation

The example's own test checks the self-field arithmetic (a 1 kg point mass at 1 m shifts clocks by G/c² = 7.4261×10^{-28}, and a disk's J is ½MR²ω) and that the program runs. The physics is held in *spacetime/*:

test	check	held to
TestGravityProbeB	geodetic 6,606.1 and frame dragging 39.2 mas/yr (GR, 642 km polar orbit)	0.5 % and 3 % (spherical Earth, no J2); within 2σ of the measured 6,601.8 ± 18.3 and 37.2 ± 7.2

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test	check	held to
TestRimRateTwin	exact rim rate = weak-field first-order rate	second order in the small quantities
TestSagnacEarthGyro	1 m² loop on the ground at 45°: $4A\omega \sin\varphi/c^2$; local-frame correction 10^{-11} to 10^{-8} of it	1e-12
TestGPSClock	GPS clock net +38.6 μs/day against the rotating Earth (model 38.54 without J2); gravitational part 45.7	Ashby 2003
TestPoundRebka	gh/c^2 over 22.5 m = 2.45×10^{-15}	Pound and Snider 1965
TestShapiroDelay, TestSolarDeflection	closed form against the Plebanski index path integral; solar-limb deflection 1.751″	Shapiro 1964; Will 2014
TestHorizonAndOrbitErrors, TestTidal	$r = 3r_s$ static rate $\sqrt{(2/3)}$; typed errors at the horizon and inside the photon sphere; Earth-surface tidal gradient 3.08×10^{-6} s^{-2}	exact
TestRotorShift, TestSagnac, TestBurstAndRotor	rotor shift with k = 1/2 equals exact rim dilation (H17's k beside it); exact Sagnac equals $4A\omega/c^2$; T1000G burst speed	exact

Limits

- Earth is spherical with no J2 term; the GP-B test tolerances (0.5 % geodetic, 3 % frame dragging) are stated with that model.
- The neutron star uses assumed values, not a measured object; in its strong field the precession and Sagnac terms are first-order and indicative only.
- The thin-hoop burst speed is an upper bound; a solid composite disk bursts lower.
- H17 (rotor Mössbauer anomaly, k ≈ 0.68 reported against the relativistic 1/2) is carried as a parameter in *spacetime.RotorShift*; this run does not print H17 values. Its register status is "deviation reported, not independently reproduced".

Reproducing the results

```
go run ./examples/machine_spacetime
go test ./examples/machine_spacetime ./spacetime
```

Sources: C. W. F. Everitt et al., PRL 106, 221101 (2011); N. Bartel et al., arXiv:1509.07529; L. I. Schiff, PRL 4, 215 (1960); N. Ashby, Living Rev. Relativ. 6, 1 (2003); Misner, Thorne and Wheeler, Gravitation (1973); J. Plebanski, Phys. Rev. 118, 1396 (1960); C. M. Will, Living Rev. Relativ. 17, 4 (2014); G. Sagnac (1913); E. J. Post, Rev. Mod. Phys. 39, 475 (1967); J. Bailey et al., Nature 268, 301 (1977); B. Mashhoon, Phys. Lett. A 145, 147 (1990); IAU 2015 Resolution B3; Toray T1000G data sheet (2024); S. Timoshenko, Strength of Materials II (1956).

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